Midterm Exam (Operating System)

答案卷請註明學號，班別，組別，姓名，以方便成績輸入.滿分:120

1. (2%)班別：[上午 A 班 | 下午 B 班]，組別：
2. F(T or F) (2%) Disk Scheduler prefers (偏愛) I/O bound process.
3. F (T or F) (2%)The fork( ) system call has at least one non-null argument.
4. T(T or F) (2%)The exec( ) system call has at least one non-null argument.
5. T(T or F) (2%)The thread_create( ) system call has at least one non-null argument.
6. F(T or F) (2%) Local variables are shared among sibling threads.
7. A (2%) Mechanism determines (a) how to do something (b) what will be done
8. (4%) Please state two factors that related to the learning effect of watching an instruction video clip. Please summarize these factors into an OS terminology. 同一個學生，同一個教學影片，觀賞影片後的學習成效也不是始終固定不變，(a) 請舉出兩項影響學習成效的因素?疲倦，飢餓，有沒有預習，… (b) 請歸納到 OS 的一個術語 context
9. (4%) What is the purpose of the command interpreter? Why is it usually separate from the kernel? command interpreter 的用途是什麼？為什麼它通常都不在核心之中？命令直譯器讀取使用者輸入的命令或者讀取命令檔，解讀並且執行之。通常都在核心之外，因為命令的格式，意義都不是一成不變。如 Linux 作業系統，更換 shell 是很平常的事情。這樣的工作，並不需要放在核心之中。
10. (4%) Two processes without parent-child relationship cannot use ordinary pipe to communicate the other. (a) Please state the reason for the restriction of parent-child. The Child process can easily find the pipe created by the Father process. (b) If they do need, what kind of pipe can be used in this situation? Named Pipe
11. (4%)(a) What is the benefit that "a process is restricted to execute on a particular processor" ?請說明「將行程限定在只能由某個特定的 processor 執行」能帶來什麼好處. 降低 cache 的 missing 以及 reload，降低 context switch 時間 (b) Please summarize above words into an OS terminology.前述文字的意義等同於哪一個 OS 專有名詞? Processor Affinity
12. (5%) SJF scheduling 有兩種版本，preemptive 以及 non-preemptive 的，FCFS 是否也有兩種版本？NO, priority scheduling 是否也有兩種版本？YES 說明理由。FCFS 以到達時間為優先序，晚到的 priority 較低，永遠不會 preemptive
13. (5%) Process A issued a disk-read I/O operation, and its state was changed to “waiting”. Operating system choose Process B to execute, and priority of B is higher than Process A. Suppose that the disk operation of Process A is finished, and an interrupt signal is raised. (a) CPU is going to run the ISR or not? 關鍵在於行程 B 的優先序是否高於硬碟的中斷(而非行程 A)，通常答案為否，所以會執行 ISR，反之，則行程 B 需要遮罩硬碟的中斷，以保證其繼續執行(b) What will be the state of Process A?前面的答案會決定何時執行 ISR，執行 ISR 之前行程 A 的狀態維持 waiting，執行 ISR 之後，行程 A 的狀態才會轉為 ready
14. (5%) The benefits of the microkernel approach include “make extending the operating system easier” and “provide more security”. Explain the reasons.
15. (5%) In the taxonomy (分類學) of interrupt, what are the similarity and differences between “Processor-detected exceptions” and “Programmed exceptions”? 這兩種 exceptions 的共同點是什麼? 都是同步中斷，也就都是由 CPU 執行指令來觸發的，相異點又是什麼? Processor-detected exceptions 是一般的指令，但執行過程中，發生了特別的事件，所以是偶然發生 exception，而 Programmed exceptions 是特別的指令，例如呼叫系統呼叫，所以是必然發生 exception
16. (5%) Interrupt Latency and Dispatch Latency are two types of latencies(延遲) which affect the performance(效能) of real-time system. (a)Please describe these two latencies, (b)Explain how they affect the performance of real-time system. 解釋這兩種延遲,它們如何影響即時系統效能
17. (5%) What is the fundamental knowledge of a Junior student of

Computer Science in “the usage of thread”? 資工系大三學生對於”thread 的使用”應該具備那些知識?

18. (10%) How many processes including the original process will the following C programs with the unix system call fork() create? Suppose all fork() system calls succeed.

```
int main(){
  int i;
  for(i = 0;i < 3; i++) {
    if(fork( ) == 0) fork( );
  }
}
```

3*3*3=27

19. (10%) (a)Please define preemptive and non-preemptive CPU scheduling. (b) Which one is preferred by modern Operating System? Explain your reasons. (c) 請詳細解釋 preemptive scheduling 與 non-preemptive scheduling 發生的時機及其發生的原因

Non-preemptive ：正在執行的行程，它的 CPU 使用權利無法被剝奪，除非它自願讓出 CPU。反之 preemptive 就是 OS 可以強行將正在使用 CPU 的行程踢出。

例如 A 行程正在執行，此時優先權較高 B 的行程進入系統，Non-preemptive 的系統會繼續執行 A，而 Preemptive 則會將 A 踢走，改執行 B。

Non-preemptive 的好處是 context switch 的次數較少，節省時間。壞處則是敏感度不夠，無法即時滿足優先權高的工作。

目前的作業系統大多是 preemptive，它比較能夠及時反應外界的需求。透過 round-robin 的 time quantum 設計，作業系統更能夠掌控電腦的行為。

non-preemptive scheduling: running □ waiting, running □ terminated, 沒有行程在 running

preemptive scheduling: new □ 前面的兩項以及 ready, waiting □ ready, running □ ready, 有新的行程到 ready

20. (20%) There are three queues (Q0, Q1, and Q2) in the system, the priority of Q0 is the highest, and then Q1 and Q2, whose priority is the lowest. The scheduling of Q0 and Q1 are RR, with time quantum 3(Q0) and 6(Q1), whereas Q2 follows the Shortest Remaining Time strategy. When a process is ready to run, it is moved into Q0. Exhausting time-quantum demotes a process to a lower priority queue, i.e., Q0 to Q1 and Q1 to Q2. Assume all jobs arrive at time zero with the following execution times (in ms): 7, 2, 20, 13, and 14. Please draw a Gantt chart to demonstrate the selection of queues as well as processes. Calculate the waiting time for each process. 畫出 Gantt chart，圖中需標明 executing process 以及其所屬的 queue。算出每一個行程的等待時間。

0	3	5	8	11	14	18	24	30	36	40	45	56		
A	B	C	D	E	A	C	D	E	D	E	C			
Q0	Q0	Q0	Q0	Q0	Q1	Q1	Q1	Q1	Q2	Q2	Q2			

Process	Arrival	End	Execution	Turnaround	Waiting
A	0	18	7	18	11
B	0	5	2	5	3
C	0	56	20	56	36
D	0	40	13	40	27
E	0	45	14	45	31

21. (20%) Explain the following terms. Please demonstrate your knowledge in associate with those terms, instead of translations. Examples are very welcome. Clear description helps credit. (a)daemon process (b)system programs (c)thread control block (d) system call

解釋下列名詞。請展現對於作業系統的理解程度，切勿翻譯了事。盡可能多用例子。字數太少，資訊不足都很可能導致無法正確的評估。因此，請盡量多寫，寫清楚。惜墨如金的，責任自負。